Topic 1 — Indian Physical Geography: Mountains & Hills

> **Sources baked in:** NCERT Geography Class 11 (Ch 2–3), Class 12 (Ch 1–2), physiography map (Kullar), UPPCS Prelims PYQs 2018–2025 **Weight:** ★★★ — map matching, peaks, passes, physiographic traps every year **Last verified:** August 2026 **Current Affairs:** Sela Tunnel (Mar 2024, Arunachal); Z-Morh / Sonamarg Tunnel (2025); Atal Tunnel = longest highway tunnel **above 10,000 ft** (not unqualified "world's longest" —)

Current Affairs (this topic)

Year	Fact	Why asked	Source
Mar 2024	**Sela Tunnel** (BRO) — all-weather Tezpur–**Tawang** (Arunachal); longest twin-lane tunnel **above ~13,000 ft**	Border tunnel / pass CA	PIB / inauguration
2025	**Z-Morh / Sonamarg Tunnel** — all-weather Srinagar–Ladakh (लद्दाख) axis	Pair with Zoji La (ज़ोजिला)	PIB
2020 / PYQ 2025	Atal Tunnel qualifier only: longest highway tunnel **above 10,000 ft**, not unqualified world longest		PIB / BRO

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q84

The Shiwalik range is primarily composed of which type of material?

A. Igneous rocks B. Consolidated rocks C. Unconsolidated sediments D. Metamorphic rocks

Show answer

Logic: Youth of the Shiwaliks explains unconsolidated sediment fill. **Ans: C.** Shiwaliks are thick unconsolidated sediments from rising Himalaya (हिमालय) — landslide-prone and dun-forming.

Q. UKPCS Prelims 2025, Q90

Which is the largest glacier of the Trans-Himalayas?

A. Biafo B. Siachen (सियाचिन) C. Baltoro D. Hispar

Show answer

Logic: Among Karakoram glaciers, relative (सापेक्ष) length decides it. **Ans: B.** Siachen (~76 km) is the largest Trans-Himalayan / Karakoram glacier among the options.

1.0 Physiographic framework

- NCERT splits India's geology into **three** structural (संरचनात्मक) units. Standard relief maps then teach **six** relief divisions on that skeleton.

Structural unit	What it is	Note
Peninsular Block	Oldest stable crust; gneiss, granite, schist	Faulted and eroded, **not** recently folded like the Himalaya
Himalayas and extra-peninsular mountains	Young fold belt from India–Eurasia collision	Still rising in zones; includes Karakoram and Purvanchal
Indo–Ganga–Brahmaputra (ब्रह्मपुत्र) Plain	Foredeep filled with Himalayan alluvium	One of the world's largest alluvial (जलोढ़) tracts

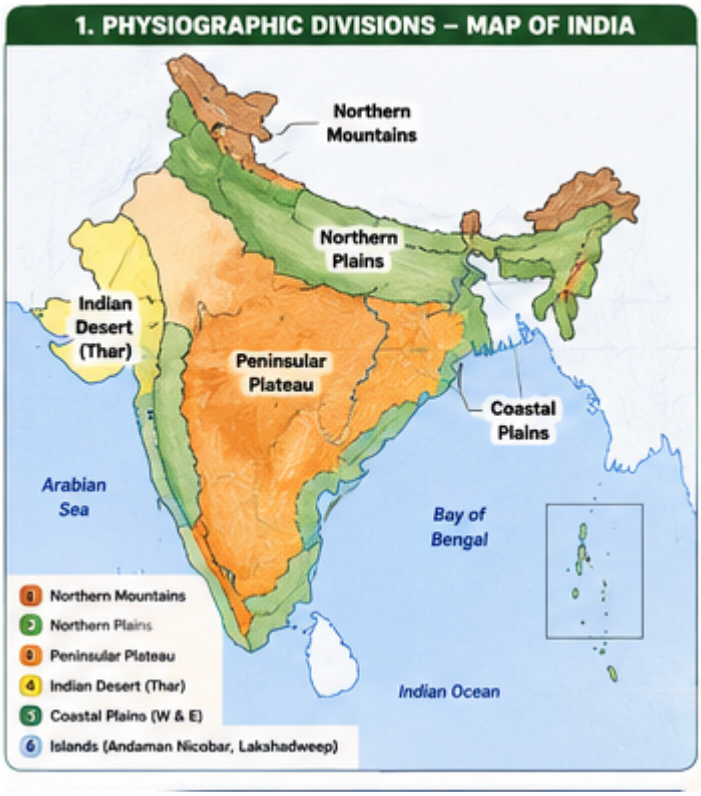


- The **six** relief units used in map questions are Northern Mountains, Northern Plains, Peninsular Plateau, Indian Desert, Coastal Plains, and Islands.
- Some notes list only **five** units and fold the Thar into the **Rajasthan Plain**. Keep the Thar as a **separate** Pleistocene–recent sand sheet, so use six divisions.
- About **43%** of India is plains, **28%** plateau, **19%** hills, and **11%** high mountains (standard class figures).

- The Himalaya is a **climatic wall**. It blocks cold Central Asian winds in winter and forces orographic rain on the southern slopes in the monsoon.
- The Peninsular Plateau is **not** the same as the Deccan Plateau. The Deccan is the **southern** tableland. The Central Highlands (Malwa, Bundelkhand, Baghelkhand) are the **northern** part of the same old block.
- The plateau is tilted **high in the west and low in the east**. That is why most peninsular rivers drain to the Bay of Bengal.
- The plateau is separated from the Northern Plains roughly along the **Narmada–Son** line.
- The **Malda Gap** (Rajmahal–Garo gap) separates the Meghalaya Plateau from the main peninsular block.

> **Logic:** Thar sands = **Pleistocene and recent** (not Paleocene/Oligocene/Pliocene). Gujarat = longest **state** coastline (~1600 km mainland).

1.1 Physiographic Divisions of India



Division	Where	Age	One-line identity
Northern Mountains	Northern frontier	Tertiary (young)	Himalaya + Karakoram + Purvanchal
Northern Plains	Between Himalaya and plateau	Recent	Indus–Ganga–Brahmaputra alluvium
Peninsular Plateau	South of Narmada–Son	Precambrian	Oldest stable block

Division	Where	Age	One-line identity
Indian Desert	Western Rajasthan	Pleistocene–recent	Thar / Marusthali (मरुस्थली), leeward of Aravalli
Coastal Plains	East and west margins	Recent	West **narrow**; East **broad and deltaic**
Islands	Arabian Sea and Bay of Bengal	Recent	Lakshadweep **coral**; A&N **volcanic**

PYQ — UPPCS Prelims 2018, Q101

Rajasthan desert or Thar desert is the expanse of which of the following?

A. Pliocene

B. Paleocene

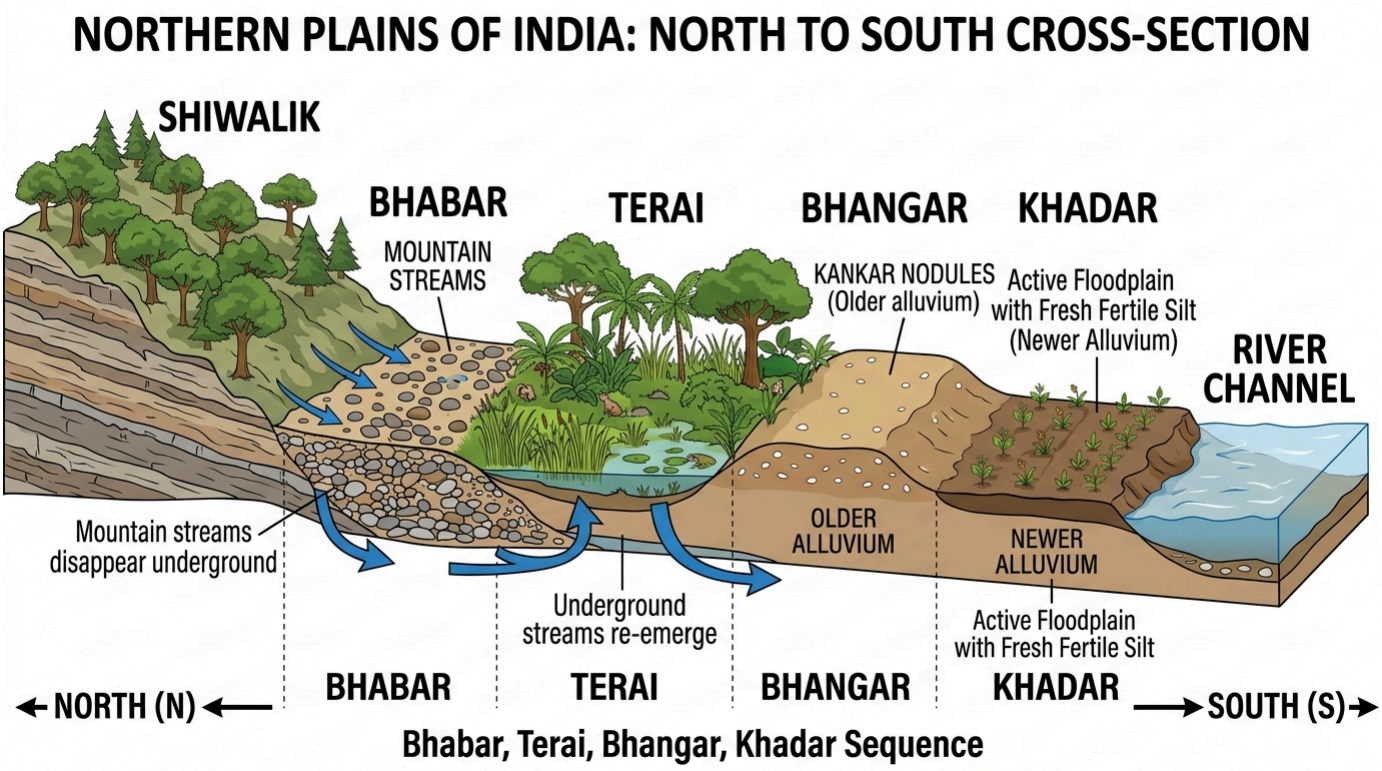
C. Pleistocene and recent deposits

D. Oligocene

Show answer

Ans: C — Quaternary sands. A/B/D are older Tertiary labels used as traps.

Northern Plains



- The Northern Plain is a **foredeep** filled by Himalayan rivers. South of the Shiwaliks the belts run **Bhabar / Bhabhar** (भाबर), then **Terai / Tarai** (तराई), then **Bhangar (भांगर) / Khadar (खादर)**. Never reverse that order.

- Direction matters. From the **Himalaya foot moving south**, the order is **Shiwalik** → **Bhabar/Bhabhar** → **Terai/Tarai**. From the **Gangetic edge moving north**, UK-style lists read **Tarai** → **Bhabar** → **Shiwalik**. Same belts — opposite reading (रीडिंग) direction. Always note where the stem starts.
- The Indian sector of this alluvial tract is about **2400 km** long. Width is greatest in the west (about **500 km**) and narrows eastward.
- Average elevation is about **200 m**. The **Ambala** watershed (about **291 m**) divides the **Indus** (सिंधु) and **Ganga** (गंगा) systems.
- The **Bhabar (Bhabhar)** is a narrow pebble belt, about **8–16 km** wide, immediately south of the Shiwaliks. Streams **sink** into the porous gravel here.
- Bhabar fans run with remarkable continuity from the **Indus to the Tista**. The belt is wider in the west than in the east.
- Bhabar is **poor for plough agriculture**. Only deep-rooted trees do well.
- The **Terai (Tarai)** lies south of the Bhabar. Streams **re-emerge**, so the belt is marshy, waterlogged, and forested. It is about **15–30 km** wide. Characteristic **artesian wells** (pressurized groundwater rising to surface naturally without pumping) are prominently found in this Tarai region (e.g., Uttarakhand and UP Terai belts).
- Terai is **more marked in the east** because rainfall is higher there.
- **Jim Corbett** (जिम कॉर्बेट) (Uttarakhand) and **Kaziranga** (काजीरंगा) (Assam) sit in terai / floodplain forest.
- **Dudhwa** (दुधवा) National Park and **Katarniaghat** Wildlife Sanctuary sit in the Terai of Uttar Pradesh.
- **Bhangar** (भांगर) is older alluvium above the present flood level. It carries **kankar** (calcareous nodules) and is less fertile.
- **Barind** in the Bengal delta (डेल्टा) and **bhur** (wind-blown sand mounds in the Ganga–Yamuna Doab) are regional names of older alluvium.
- **Khadar** (खादर) is newer floodplain alluvium, renewed almost every flood, and is **more fertile** than Bhangar.
- Alkaline patches on older alluvium are called **reh** or **kallar** (also **kollar** in drier Haryana). Extra irrigation can spread reh by capillary salts.

Sub-region	Location	Fact
Punjab Plain	Punjab, Haryana	Doabs; **chos** gullying on Shiwalik edge
Haryana Tract	Ghaggar–Yamuna	Water-divide; **Ghaggar** (घग्गर) as Saraswati (सरस्वती) successor
Ganga Plain	Delhi to Kolkata	Rohilkhand, Awadh (अवध), Mithila, Magadh
Brahmaputra Plain	Assam	Braided channel with **bils** (ox-bows) and heavy floods

- A **doab** (दोआब) is the land between two rivers.

Doab	Between
Bist	Beas (ब्यास) and Satluj

Doab	Between
Bari	Beas and Ravi (रावी)
Rachna	Ravi and Chenab (चिनाब)
Chaj	Chenab and Jhelum (झेलम)
Sind (सिंध) Sagar	Jhelum–Chenab and the Indus

- The **Ganga–Yamuna Doab** is the core agricultural heart of western Uttar Pradesh.
- **Majuli** in Assam is the world's largest inhabited **river island** (shrinking by erosion). **Kaziranga** sits on the Brahmaputra floodplain.
- The Ganga delta is **arcuate** (चापाकार). The **Sundarbans** (सुंदरबन) mangrove tract straddles India and Bangladesh.

> **Logic:** Khadar is newer and more fertile. Never swap it with Bhangar. Duars of West Bengal–Assam are a **different** foothill tract from the UP Terai.

Indian Desert (Thar / Marusthali)

- The Thar lies on the **leeward** side of the Aravalli. Core rainfall is generally **below 250 mm**.
- At depth the desert sits on **peninsular** gneiss and granite. Only the surface looks like a young sand sheet.
- **Marusthali** is the sandy desert core with shifting **barchans**. The **west** is sandier; the **east** is rockier.
- **Bagar** (बागड़) is the semi-arid eastern fringe up to the Aravalli, with short seasonal streams and **rohi** patches.
- The tract **north of the Luni (लूनी)** is the **thali** (sandy plain) with inland saline (लवणीय) lakes.
- **Sambhar Lake** near Jaipur is the largest inland salt (लवणाध्यक्ष) lake. **Didwana, Degana, and Kuchaman** are other saline lakes.
- The **Luni** (लूनी) is the only major river of the region. It is **seasonal** and **inland-draining**. It is not a perennial Arabian Sea river.
- The southern margin is the **Rann of Kachchh**. The eastern margin is the Aravalli.

> **Logic:** Thar sands = Pleistocene and recent (not older Tertiary epochs).

Coastal Plains

- The **Western Coastal Plain** is **narrow** (about 50–80 km) because the Western Ghats rise close to the Arabian Sea.
- The **Eastern Coastal Plain** is **broad** (about 100–200 km) and **deltaic**. Average width is about **120 km**.
- India's mainland coast is fairly **straight** (Cretaceous Gondwana faulting). It offers **fewer** deep natural harbours than an indented European coast.

- **Konkan** (Maharashtra–Goa (गोवा)) is a coast of **submergence** (faulted, drowned inlets, Mumbai harbour).
- **Malabar / Kerala** is a coast of **emergence** (lagoons, spits, kayals).
- **Coromandel** (Tamil Nadu) is a coast of **emergence**. Deltas silt harbours.
- **Kachchh and Kathiawar** are geologically a **peninsular** extension (Kathiawar has Deccan lava). They are still mapped with the western coastal plains because they are now levelled.
- **Kachchh** was once an island. The **Great Rann** is the salt waste to its north. The **Little Rann** lies to the south-east.
- **Girnar (1117 m)** is a volcanic high on Kathiawar. The **Gir** (गीर) forest is the last home of the Asiatic lion (सिंह).

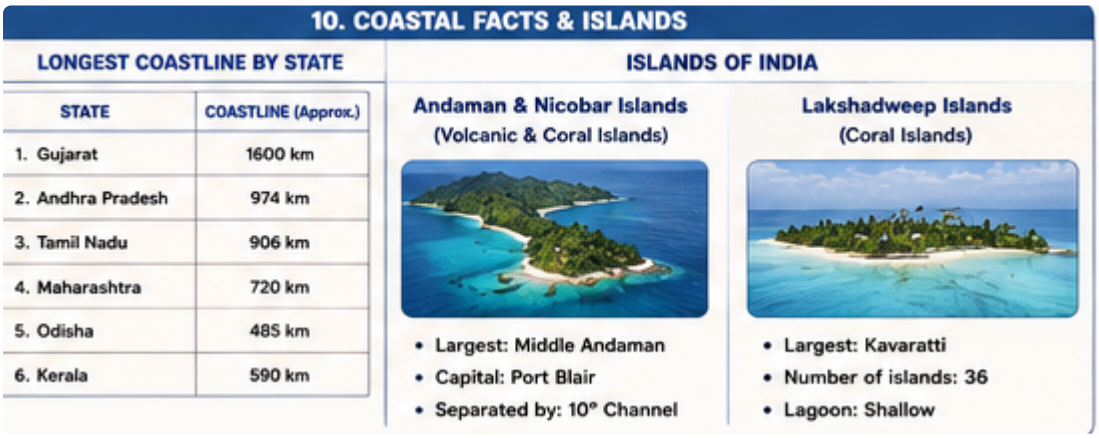
Western strip	Stretch
Kachchh–Kathiawar	Gujarat
Gujarat Plain	Narmada–Tapi–Mahi (माही)–Sabarmati (साबरमती) fill
Konkan	Daman to Goa
Kanara (Canara)	Goa to Mangaluru
Malabar	Kerala
Eastern strip	Stretch
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Utkal Coast	Odisha coast with the Mahanadi (महानदी) delta and **Chilika**
Northern Circars	Mahanadi to Krishna (कृष्णा)
Carnatic (कर्नाटक)	Krishna to Kaveri (कावेरी)
Coromandel / Payan Ghat	False Divi Point to Kanyakumari

- The **Sharavati** drops as **Jog (Gersoppa) Falls** (about 271 m) on the Karnataka Ghats scarp.
- **Sriharikota** is the sand spit that bars **Pulicat** Lake (ISRO).
- **Kolleru** was once a coastal lagoon. Emergence has left it **inland** on the combined Godavari (गोदावरी)–Krishna delta.
- The **Kaveri delta** (about 130 km wide) is the "granary of South India."
- The **Kerala kayals** (backwaters) are lagoon–estuary (ज्वारनदमुख) waterways. **Vembanad** is the largest.
- **Chilika** (Odisha) is India's largest coastal lagoon.
- Kerala beach sands carry **monazite** (thorium). The KG offshore basin is a hydrocarbon play.
- West-flowing Narmada and Tapi form **estuaries**, not large deltas.

> **Logic:** Do not reverse widths. West is narrow; east is broad. Do not call the **whole** west coast submerged — only **Konkan** is the classic drowned coast.

Islands of India

- The **Andaman and Nicobar** chain is a southward continuation of the **Arakan Yoma** (Purvanchal–Myanmar arc). The rocks are **volcanic / tectonic**. The capital is **Port (बंदरगाह) Blair** on **South Andaman**.
- **Saddle Peak (737 m)** on **North Andaman** is the highest point of the archipelago.
- **Barren Island** is India's **only active volcano**. **Narcondam** is volcanic but **dormant / extinct**. (Do not copy the older "both active" line.)
- The **Ten Degree Channel** separates the Andaman group from the Nicobar group.
- **Duncan Passage** separates **South Andaman** from **Little Andaman**.
- The **Coco Channel** lies between North Andaman and Myanmar's Coco Islands.
- **Car Nicobar** is the northernmost Nicobar island. **Great Nicobar** is the largest and southernmost, close to **Sumatra**.
- **Indira Point** on **Great Nicobar** (about 6°45′ N) is India's southernmost **territory**. It is not Kanyakumari.
- **Lakshadweep** (लक्षद्वीप) islands are **coral atolls** (Reunion hotspot trail) about 200–500 km off Kerala. The capital is **Kavaratti**.
- The old three-fold names are **Amindivi** (north), **Laccadive** (middle), and **Minicoy** (south).
- **Minicoy** is the largest and southernmost inhabited island of Lakshadweep. Elevation is generally **under 5 m**, so the group is highly exposed to sea-level rise.
- The **Nine Degree Channel** separates **Minicoy** from the rest of Lakshadweep.
- The **Eight Degree Channel** separates **Minicoy** from the **Maldives**.
- Adam's Bridge / **Rama Setu** remnants lie between India and Sri Lanka. They are a submerged limestone shoal, not a Himalayan fold.



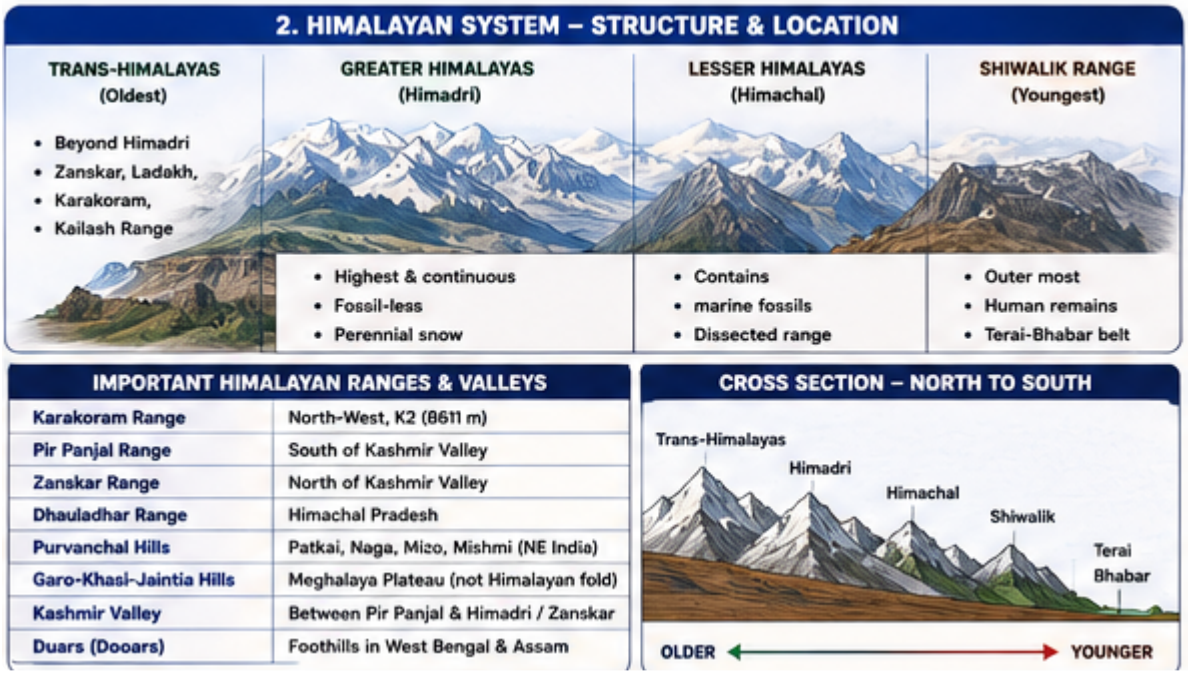
> **Logic:** Never reverse origins. A&N = volcanic. Lakshadweep = coral.

1.2 Himalayan Mountain System

- The Himalaya is a **young fold** belt from the **Indian–Eurasian** collision (about 50 million years ago). The arc is about **2400 km** from the **Indus gorge** (west) to the **Dihang gorge** (east).

- The arc has two **syntaxial bends**. The western bend is around **Nanga Parbat**. The eastern bend is around **Namcha Barwa** in **Tibet**, not in India.

North to south (parallel ranges)



Range	Typical height	Rock / fossil fact
Trans-Himalaya	3000–6500 m	Granitic cold desert **north of the Indus**
Himadri (Greater)	generally **>6000 m**	Crystalline, **fossil-less**, with permanent snow
Himachal (हिमाचल) (Lesser)	3700–4500 m	Sedimentary; **marine fossils**
Shiwalik (शिवालिक) (Outer)	900–1100 m	Youngest unconsolidated rocks; **human remains**

- The **Himadri** is the highest and most continuous range. It is crystalline (granite–gneiss) with a steep **south** face and gentler **north** face (**hogback**).
- The **Himachal** belt holds most famous hill stations. **Pir Panjal**, **Dhauladhar** (धौलाधार), **Mussoorie** (मसूरी), and **Nag Tibba** belong here.
- **Mahabharat Lekh** is the Lesser Himalaya name in **Nepal**.
- The **Shiwalik** is the outermost and youngest range (about 600–1500 m; often quoted 900–1100 m). It is folded conglomerate fans, not crystalline rock.
- Seasonal torrents that dissect the Punjab–Himachal Shiwalik are **chos**. Forest cover (वन आवरण) on Shiwalik **decreases westwards** with rainfall.

Region	Local name of Shiwalik
Jammu	Jammu Hills
Uttarakhand	Dhang / Dundwa
Nepal	Churia Ghat Hills

Region	Local name of Shiwalik
Arunachal	Dafla, Miri (मीरी), Abor, Mishmi (outer hills)

- Longitudinal valleys between Himachal and Shiwalik are **duns** in the west and **duars** in the east.
- **Dehra Dun** is the best-known dun (about 75 km by 15–20 km). **Kotli, Patli, Kyarda, Chaukhamba (चौखंबा), and Udhampur** are other named duns.
- High snow-covered ranges feed **perennial** Himalayan rivers (Indus, Ganga, Brahmaputra headwaters).
- Altitude creates climate belts, so vegetation changes from tropical through temperate and alpine to nival.

> **Logic:** All three fossil facts are true. Greater Himalaya is **fossil-less**. Lesser Himalaya carries **marine fossils**. Shiwalik carries **human remains**. Among usual options the **youngest** range is **Himadri** (Aravalli is oldest). Snow-melt feeds perennial rivers. Altitude creates vegetation zonation.

PYQ — UPPCS Prelims 2019, Q11

With reference to the Himalayan range, which of the statements is/are correct?

1. The sedimentary rocks of the greater Himalayas were fossil less.
2. Marine livings fossils are found in the sedimentary rocks of lesser Himalayas.
3. Remains of human civilization are found in outer or Shivalik (शिवालिक) Himalayas.

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3 are correct

Show answer

Ans: D — All three zonation facts. Trap: putting marine fossils in Greater Himalaya.

West to east (regional sections)



1. Longitudinal River Divisions (West → East by Sir Sidney Burrard)

Section (खंड)	River Boundaries (नदी सीमाएं)	Length (लंबाई)	Key Ranges & Peaks (प्रमुख श्रेणियां व चोटियां)
Punjab / Kashmir / Himachal Himalaya	**Indus (सिंधु) to Satluj (सतलज)**	~560 km	Karakoram, Ladakh, Zanskar, Pir Panjal, Dhauladhar; Nanga Parbat
Kumaon Himalaya (कुमाऊँ)	**Satluj (सतलज) to Kali (काली / शारदा)**	~320 km	Nanda Devi, Kamet, Badrinath, Kedarnath, Trishul
Nepal / Central Himalaya (नेपाल)	**Kali (काली) to Teesta (तीस्ता)**	~800 km *(Longest & Highest)*	Mt. Everest, Kanchenjunga, Dhaulagiri, Annapurna, Makalu
Assam / Eastern Himalaya (असम)	**Teesta (तीस्ता) to Dihang (दिहांग / ब्रह्मपुत्र)**	~720 km	Namcha Barwa, Kula Kangri, Kangto

- **Western Himalaya** as a major regional block runs **Indus to Kali** (Kashmir + Himachal + Kumaon).
- Western Himalaya rises **gradually** through multiple parallel ranges; Eastern Himalaya rises **abruptly** from the Bengal–Assam plains.
- **Sandakphu** on the **Singalila** ridge is West Bengal's highest point.
- Beyond the Dihang gorge the ranges swing sharply southwards as the **Purvanchal (पूर्वांचल)** hills.

2. Cardinal Orientation Master Guides (Exam Fact-Locks)

A. North to South Sequence of Northern Ranges (उत्तर से दक्षिण क्रम)

> **Mnemonic: K – L – Z – P – D – S** (*"Kal Ladakh Zaskar se Pir Dhaula Shiwalik"*): 1. **Karakoram Range (काराकोरम श्रेणी)** — Trans-Himalayas; K2 (Godwin Austen, 8611 m), Siachen glacier. 2. **Ladakh Range (लद्दाख श्रेणी)** — Trans-Himalayas; sits between Indus and Shyok rivers; Khardung La pass. 3. **Zaskar / Zanskar Range (जांस्कर श्रेणी)** — Trans-Himalayas / Greater Himalaya transition; separates Ladakh from Kashmir. 4. **Great Himalayas / Himadri (वृहद हिमालय / हिमाद्रि)** — Highest continuous arc; average elevation ~6,000 m. 5. **Pir Panjal Range (पीर पंजाल श्रेणी)** — Middle/Lesser Himalaya (longest range); south wall of Kashmir valley; Rohtang & Banihal passes. 6. **Dhauladhar Range (धौलाधार श्रेणी)** — Middle/Lesser Himalaya in Himachal Pradesh; Kangra valley at foot. 7. **Shiwalik Range (शिवालिक श्रेणी)** — Outer/Sub-Himalaya; youngest sedimentary fold; elevation 900–1100 m.

B. West to East Sequence of Major Himalayan Peaks (पश्चिम से पूर्व चोटियां)

> **West → East: Nanga Parbat (8126 m, PoK/J&K) → Kamet (7756 m, Uttarakhand) → Nanda Devi (7816 m, Uttarakhand) → Dhaulagiri (8167 m, Nepal) → Annapurna (8091 m, Nepal) → Manaslu (8163 m, Nepal) → Cho Oyu (8188 m, Nepal) → Mt. Everest (8848.86 m, Nepal/Tibet) → Lhotse (8516 m, Nepal) → Makalu (8485 m, Nepal) → Kanchenjunga (8586 m, Sikkim) → Namcha Barwa (7782 m, Tibet/Arunachal)**

C. West to East Sequence of Arunachal / Purvanchal Foothills

> **West → East: Dafla Hills (डफला) → Miri Hills (मिरी) → Abor Hills (अबोर्) → Mishmi Hills (मिशमी)**

D. West to East Sequence of West Asian to Indian Ranges (UPPCS 2024 Paper Pattern)

> **West → East: Pontic (Turkey) → Zagros (Iran) → Hindu Kush (Afghanistan) → Karakoram (Ladakh/Pak/China)**

Trans-Himalaya, Karakoram, glaciers

- The Trans-Himalaya is the **Tibetan Himalaya** zone. The monsoon does **not** reach it because Himadri casts a rain shadow.
- The **Zaskar (Zanskar)**, **Ladakh**, and **Kailash** ranges are the Trans-Himalayan set taught for India.
- The **Kailash (Gangdise)** range is treated as an offshoot of the Ladakh Range. The **Indus** rises on its northern slope.
- The **Karakoram** (also called **Krishnagiri**) occupies the extreme north-west and joins the **Pamir Knot**.
- **K2 (Godwin Austin, 8611 m)** is in the Karakoram in **PoK**. It is **not** on the Himadri.

- North-east of Karakoram the **Ladakh Plateau** is cut into **Aksai Chin, Depsang, Lingzi Tang, Chang Chenmo, and Soda Plains**.
- **Siachen** (about **75 km**, Nubra valley) is the longest glacier in the **Indian-administered** Karakoram. **Fedchenko** in the Pamir is longer. **Hispar** is another giant Karakoram glacier.
- **Sonapani** (about 15 km, Lahaul Chandra valley) is the longest named glacier of the **Pir Panjal** set.
- The **Gangotri** (गंगोत्री) glacier (about 30 km) feeds the **Bhagirathi** (भागीरथी). The **Yamunotri** (यमुनोत्री) glacier feeds the **Yamuna** (यमुना).
- **Zemu** glacier (Sikkim) is the largest glacier of the eastern Himalaya and feeds the **Teesta**.
- Snowline is **lower in the west** (about 2500 m) than in the **east / Kumaon** (about 3500 m), mainly because western precipitation falls more as snow at higher latitude.
- On Himadri the snowline is **lower on the steep, wetter southern slope** than on the drier north.
- The **Karakoram Pass** was a historic (ऐतिहासिक) Silk Route col. Do not confuse it with **Khardung La** (खारदुंग ला) (Ladakh road pass).

Pir Panjal, Zanskar, Dhauladhar, Kashmir Valley

- The **Pir Panjal** is the longest range of the Lesser Himalaya. It is the **southern wall** of the Kashmir Valley. It runs from the Jhelum to the upper Beas (over 300 km) and contains much **volcanic** rock.
- **Banihal Pass** carries the Jammu–Srinagar highway and the rail alignment through this wall.
- The **Atal Tunnel** (about 9.02 km) runs under **Rohtang** in the Pir Panjal of **Himachal**. It links Manali with Lahaul–Spiti.
- Unqualified “world’s longest highway tunnel” is **false**. The correct qualifier is longest highway tunnel **above 10,000 ft**.
- The **Zanskar** (ज़ांस्कर) range lies **north** of the Kashmir Valley and separates it from Ladakh. **Pensi La** crosses it. **Kargil** (कारगिल) sits toward its western end.
- The **Dhauladhar** is an outer Lesser Himalayan range in **Himachal**. **Kangra (कांगड़ा) Valley** (कांगड़ा) is a **strike** (longitudinal) valley at its foot. **Kulu Valley** (कुल्लू) (upper Ravi) is a **transverse** valley.
- The Kashmir Valley is about **135 km by 32–40 km**. The **Jhelum** (झेलम) drains it through the **Baramulla gorge**. **Srinagar**, **Dal Lake**, and **Wular Lake** belong to this vale.
- The valley lies between **Pir Panjal (south)** and **Himadri (north)**.
- **Karewas** are Pleistocene **lacustrine** terraces left when the Kashmir lake drained through Baramulla. Thickness can reach about **1400 m**.
- Karewas grow **saffron**, almond, walnut, and apple. They are **not** the same landform as a dun.

> **Logic:** Kangra–Dhauladhar is a **Himachal** pairing. Do not use it for Kashmir. Atal Tunnel is in **Pir Panjal**, not Karakoram or Zanskar.

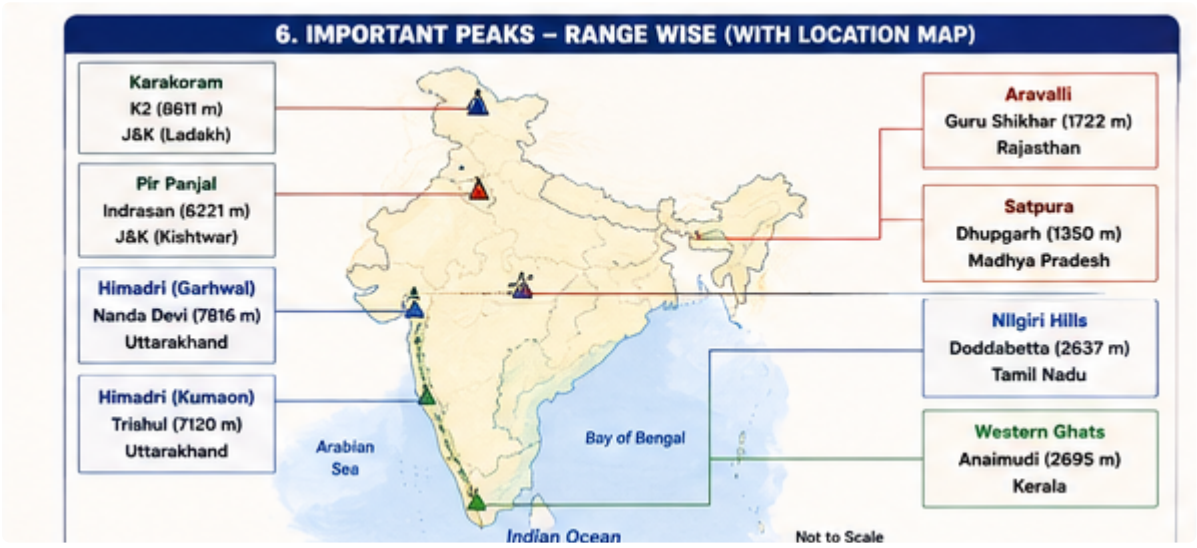
Purvanchal, Garo–Khasi–Jaintia, Duars

- **Purvanchal Hills** are the north-eastern **fold** hills beyond the Dihang. They are **convex to the west**. They are **not** eastern Uttar Pradesh.
- The **Patkai Bum** runs along the Arunachal–Myanmar border. It is mostly sandstone.
- The **Naga Hills** are in Nagaland. **Saramati (3841 m)** is Nagaland’s highest peak.

- The **Barail Range** separates the Naga Hills from the **Manipur Hills**.
- The **Mizo / Lushai Hills** are in Mizoram. **Phawngpui (Blue Mountain, 2165 m)** is Mizoram's highest.
- The **Mishmi Hills** are in eastern Arunachal.
- The **Manipur Hills** hold **Tenipu / Mount Iso**, Manipur's usual highest peak.
- **Garo, Khasi, and Jaintia** hills form the **Meghalaya Plateau**. They are **geologically peninsular**, not Himalayan folds.
- The **Garo Hills** occupy western Meghalaya.
- The **Khasi Hills** occupy central Meghalaya. **Shillong Peak (1965 m)** is Meghalaya's highest.
- **Mawsynram** and **Cherrapunji** sit on the **Khasi Hills**, not on Garo or Jaintia.
- The **Jaintia Hills** occupy eastern Meghalaya and are known for limestone caves.
- **Duars / Dooars** are the Himalayan foothills of **West Bengal and Assam**. They are the "door" to Bhutan and carry tea and wildlife corridors (Jaldapara, Gorumara, Buxa).

> **Logic:** Wettest places sit on **Khasi Hills**. Duars ≠ Terai of UP.

Himalayan peaks (match home)



Peak	m	Note
K2 (Godwin Austin)	8611	**Karakoram** (PoK) — not the main Himalayan arc
Kanchenjunga	8586	Highest peak **fully in India** (Sikkim, Himadri)
Nanda Devi	7816	Uttarakhand; highest entirely in Uttarakhand
Kamet (कामेत)	7756	Uttarakhand
Cho Oyu	8188	Himalaya (Nepal–Tibet) —
Lhotse	8516	Himalaya (Nepal–Tibet) —
Nanga Parbat	8126	Western syntaxial bend (Gilgit–Baltistan / PoK)
Namcha Barwa	7782	**Tibet / China — not** in India

Peak	m	Note
Gurla Mandhata	7694	**Tibet**, north of Mansarovar — **not** an Indian peak

- **Annamalai** and **Sirumalai** are **Tamil Nadu peninsular** hills. They are **not** Himalayan.

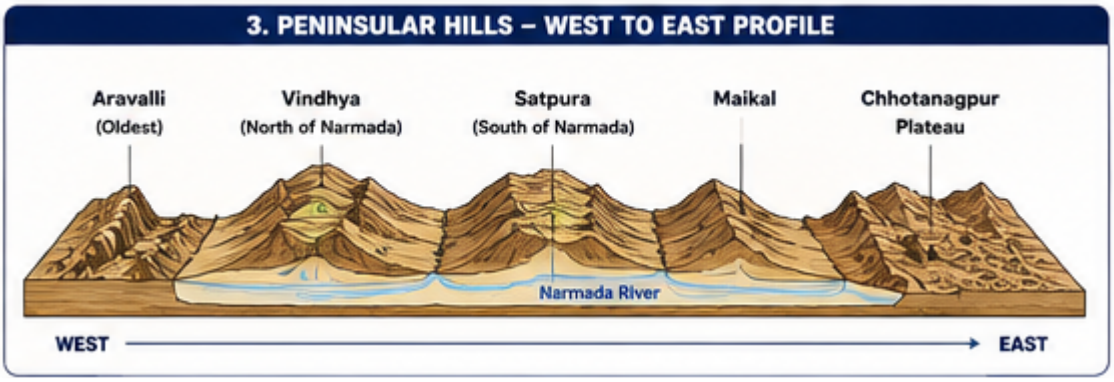
> **Logic: Namcha Barwa** is not in India. **Gurla Mandhata** is Tibetan — do not list either as an Indian Trans-Himalayan peak.

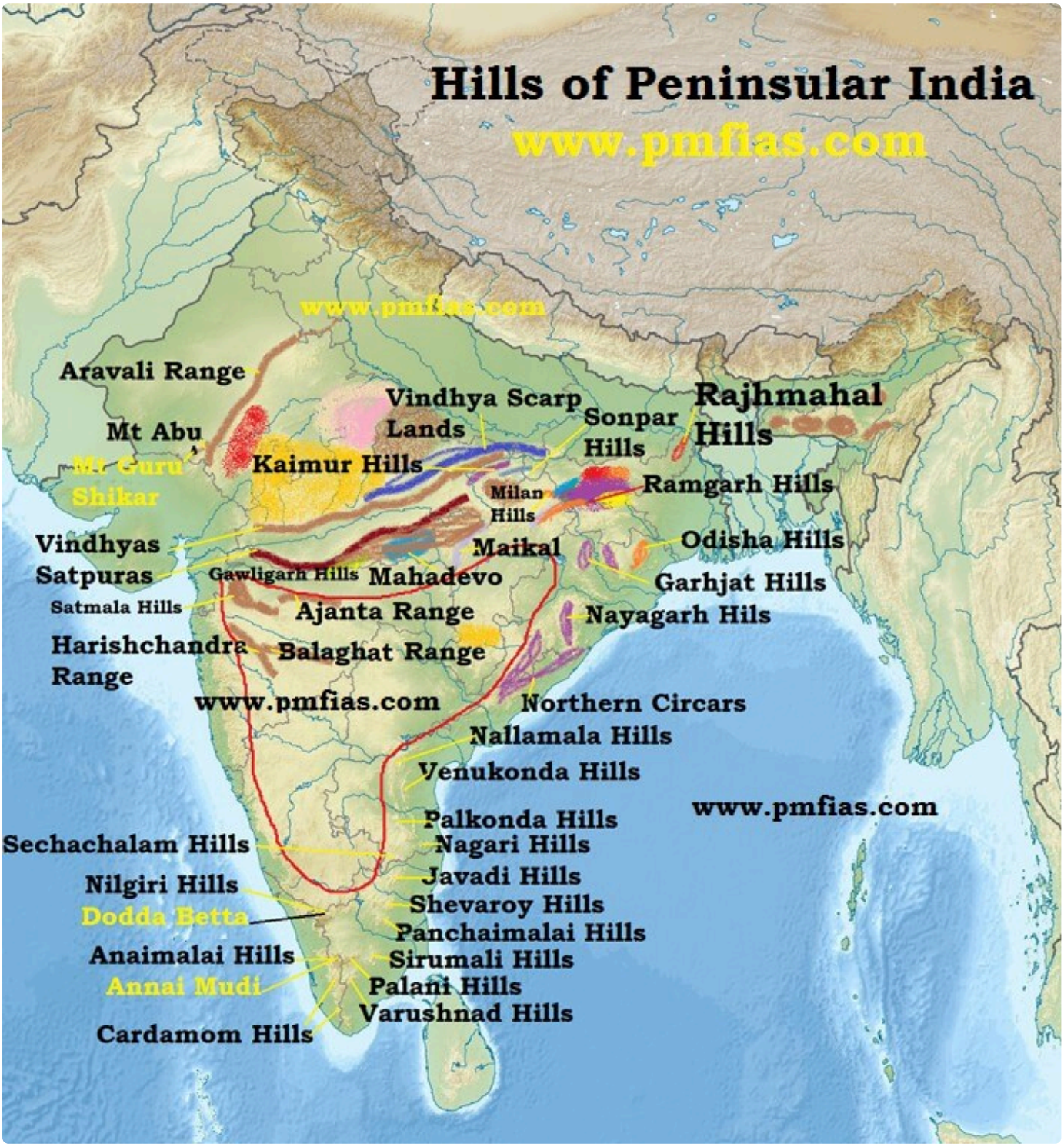
1.3 Peninsular Mountains, Hills & Plateaus

- The peninsula is an ancient **Precambrian** block. Most peninsular hills are **relict** (residual) **horsts**, not young folds.
- The plateau is roughly **triangular** (base along the northern plain, apex at Kanyakumari) and covers about **16 lakh km²**. Average height is about **600–900 m**.

Range	Trend	Location	Highest	Fact
Aravalli	SW–NE	Gujarat–Rajasthan–Delhi	**Guru Shikhar 1722 m**	**Oldest** fold mountains; Rajasthan highest
Vindhya	E–W	**North of Narmada**, MP	Sadbhawna Shikhar	Aryavarta / Dakshinapatha (दक्षिणापथ) divide
Satpura	E–W	**South of Narmada**	**Dhupgarh 1350 m** (Pachmarhi)	Madhya Pradesh highest
Western Ghats (पश्चिमी घाट)	N–S	Tapi gap to Kanyakumari	**Anaimudi 2695 m**	**Continuous**; Sahyadri in Maharashtra
Eastern Ghats	N–S	Odisha to Tamil Nadu	**Jindhagada ~1690 m**	**Discontinuous**
Nilgiri (नीलगिरि)	—	Tamil Nadu–Kerala	**Doddabetta 2637 m**	Junction of both Ghats
Rajmahal	—	Jharkhand	—	Rajmahal Trap (volcanic)
Shevaroy	—	Salem, Tamil Nadu	—	Eastern Ghats; **not** Tirupati

- Central India hills run **west to east** as **Satpura**, then **Mahadeo**, then **Maikal**, then **Chhotanagpur**.


3. PENINSULAR HILLS – WEST TO EAST PROFILE
Aravalli
(Oldest)
Vindhya
(North of Narmada)
Satpura
(South of Narmada)
Maikal
Chhotanagpur
Plateau
Narmada River
WEST
EAST


Hills of Peninsular India
www.pmfias.com
Aravali Range
Mt Abu
Mt Guru
Shikar
Vindhya Scarp
Lands
Kaimur Hills
Sonpar
Hills
Rajhmahal
Hills
Milan
Hills
Ramgarh Hills
Vindhyas
Satpuras
Satmala Hills
Gawligarh Hills
Maikal
Mahadevo
Ajanta Range
Odisha Hills
Garhjat Hills
Nayagarh Hils
Harishchandra
Range
Balaghat Range
Northern Circars
Nallamala Hills
Venukonda Hills
Palkonda Hills
Nagari Hills
Javadi Hills
Shevaroy Hills
Panchaimalai Hills
Sirumali Hills
Palani Hills
Varushnad Hills
Sechachalam Hills
Nilgiri Hills
Dodda Betta
Anaimalai Hills
Annai Mudi
Cardamom Hills

Aravalli Range

- The Aravalli is about **800 km** from **Palanpur (Gujarat)** to Delhi. It is the **oldest** fold mountain system of India (Archaean), now a **relict** range.
- **Guru Shikhar (1722 m)** on Mount Abu is Rajasthan's highest peak. Mount Abu is cut from the main range by the **Banas** valley.
- South of Ajmer (अजमेर) the range is continuous. North of Ajmer it breaks into detached ridges (Haryana–Delhi).
- The buried north-east stump continues toward **Haridwar** (हरिद्वार) under Ganga alluvium. The **Delhi Ridge** is the visible remnant.
- **Pipli Ghat, Dewair, and Desuri** are the road–rail gaps through the Rajasthan Aravalli.
- The range casts a rain shadow that helps create the **Thar** (थार).
- **Zawar** (Rajasthan) is the classic zinc–lead locality of the Aravalli belt.

Vindhya Range

- The Vindhya runs **east–west north of the Narmada** as an escarpment on the **Narmada–Son trough**, from **Jobat (Gujarat)** to **Sasaram (सासाराम) (Bihar)** (over 1200 km).
- It is the watershed between the **Ganga system** and peninsular rivers. Chambal (चंबल), Betwa (बेतवा), and Ken rise within about 30 km of the Narmada.
- The Vindhya continues east as the **Bhanrer** and **Kaimur** hills.
- **Kaimur Hills** are the Vindhyan extension into **Uttar Pradesh and Bihar**.
- **Amsot (about 941 m)** in Sonbhadra is Uttar Pradesh's highest point. It is **not** Himalayan.

Satpura Range

- The Satpura is a series of **seven** residual blocks (**sat-pura**) and is treated as a **horst**. It runs **east–west south of the Narmada**, between the Narmada and the Tapi, for about **900 km**.
- **Dhupgarh (1350 m)** at **Pachmarhi** on the **Mahadeo Hills** is Madhya Pradesh's highest peak.
- **Mahadeo Hills** continue the Satpura eastward.
- **Maikal Hills** continue still farther east into Madhya Pradesh–Chhattisgarh. **Amarkantak** sits at this eastern junction.
- **Amarkantak** is the source of the **Narmada** (नर्मदा) and the **Son**.
- The **Tapi** (तापी) flows in a rift/gorge south of the Satpura. **Gavilgarh Hills** are the western Satpura in Maharashtra.

> **Logic:** Never reverse Vindhya and Satpura across the Narmada. do not put Maikal before Satpura or Chhotanagpur before Maikal.

PYQ — UPPCS Prelims 2019, Q6

Which one of the following is the correct sequence of the hills of Central India located from West to East?

A. Maikal, Satpura, Mahadeo and Chhotanagpur

B. Satpura, Mahadeo, Maikal and Chhotanagpur

C. Maikal, Mahadeo, Satpura and Chhatanagpur

D. Satpura, Mahadeo, Chhotanagpur and Maikal

Show answer

Ans: B — Satpura → Mahadeo → Maikal → Chhotanagpur.

Western Ghats (Sahyadri)

- The Western Ghats are a **continuous** escarpment from the **Tapi valley** to near **Kanyakumari** (about **1600 km**).
- They rise as a steep wall on the west and slope **gently east** into the Deccan, so they hardly look like a mountain from the plateau.
- Northern Sahyadri is **Deccan Trap** lava with a **landing-stair** (stepped) west face.
- **Kalsubai (1646 m)** is Maharashtra's highest peak. **Salher** and **Mahabaleshwar** are other northern Sahyadri highs.
- Middle Sahyadri (south of about 16° N) is **granite–gneiss**. **Kudremukh** and **Baba Budan / Mullayanagiri** sit here.
- **Mullayanagiri (1930 m)** is Karnataka's highest peak.
- **Thal Ghat (Kasara)** carries the Mumbai–Nashik route.
- **Bhor Ghat (Khandala)** carries the Mumbai–Pune (पुणे) route.
- **Palghat (Palakkad) Gap** is a **rift** and the **widest** gap. It links Kerala with Tamil Nadu and lets south-west monsoon moisture into the Mysore (मैसूर)–Coimbatore interior.



Eastern Ghats

- The Eastern Ghats are **discontinuous** and lower. They have neither structural unity nor a continuous crest.
- True mountain character is clearest between the **Mahanadi and the Godavari** (**Maliya** and **Madugula Konda** ranges).
- **Jindhagada (about 1690 m)** in Araku Valley is the usual highest peak of the Eastern Ghats.
- **Arma Konda** (Andhra Pradesh) is another high Eastern Ghats peak used in match lists.
- **Mahendragiri (1501 m)** is the well-known Odisha peak of the Maliya range.
- Between the **Godavari and the Krishna** the Ghats almost vanish. That low is the **KG (Gondwana) basin**.
- The **Nallamala** Hills (Cuddapah–Kurnool) restore a hill chain. The southern part is the **Palkonda** Range.

- The **Javadi** Hills and the **Shevaroy–Kalrayan** Hills are the Tamil Nadu remnants.
- **Biligiri Rangan** Hills sit on the Karnataka–Tamil Nadu border.
- The **Shevaroy** Hills (Yercaud, Salem) are **not** the site of Tirupati.
- **Tirumala / Mallamalla** Hills (Tirupati) are Eastern Ghats of **Andhra Pradesh**.

Southern hills (Nilgiri–Anamalai–Cardamom)

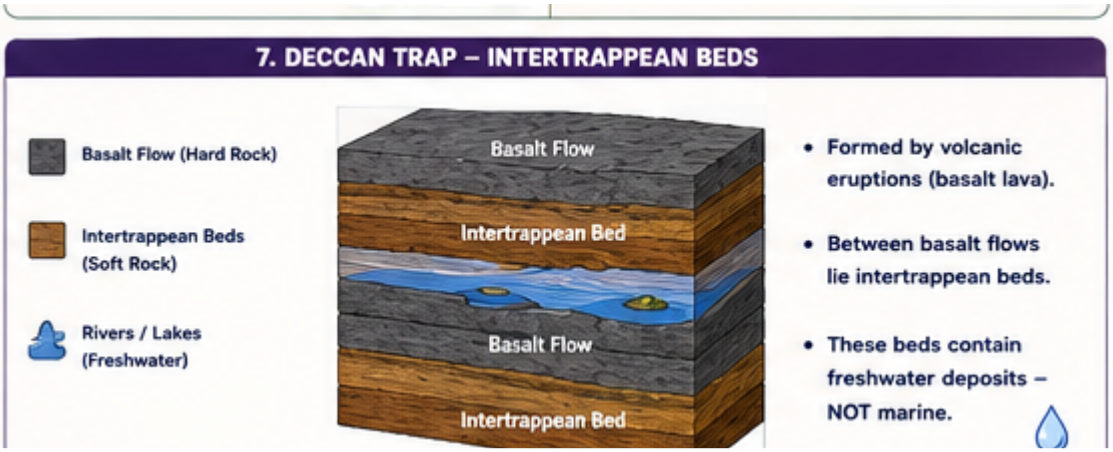
- The **Nilgiri Hills** are the junction of the Western Ghats, Eastern Ghats, and southern hills. They rise abruptly above 2000 m.
- **Doddabetta (2637 m)** is Tamil Nadu's highest peak. **Makurti** is the other named Nilgiri high.
- **Udhagamandalam (Ooty)** sits in the Nilgiris.
- South of Palghat, **Anaimudi (2695 m)** is the highest peak of the Western Ghats **and of South India**.
- Three ranges radiate from Anaimudi. The **Anamalai** runs north. The **Palani** runs north-east (**Kodaikanal**). The **Cardamom / Ealaimalai** runs south.
- **Sirumalai** is a Tamil Nadu outlier. Pair it with Annamalai as **peninsular**, not Himalayan.
- **Eravikulam** (Nilgiri Tahr) and **Periyar** (पेरियार) belong to this southern Western Ghats complex.

> **Logic:** Doddabetta = **Tamil Nadu**. Anaimudi = **Kerala**. Annamalai / Sirumalai are **not** Himalayan.

Deccan Plateau and Deccan Trap

- The **Deccan Plateau** is the **southern** triangular tableland (about **5 lakh km²**). Satpura–Vindhya bound it on the north-west, Mahadeo–Maikal on the north, Western Ghats on the west, Eastern Ghats on the east. It **slopes eastward**. Height is about **1000 m** in the south and about **500 m** in the north.
- The **Maharashtra Plateau** is trap country with **step topography** and **regur** (रेगुर).
- The **Karnataka (Mysore) Plateau** splits into **Malnad** (मलनाड) (forested hill country) and **Maidan** (मैदान) (rolling granite plain). It tapers into the Nilgiris.
- The **Telangana Plateau** is Archaean gneiss drained by Godavari, Krishna, and Penner.
- The **Chhattisgarh Plain** is the only true plain inside the peninsula: a **saucer** of limestone–shale drained by the upper **Mahanadi** (महानदी), between Maikal and the Odisha hills.
- The **Deccan Trap** is a **basalt** cover from fissure eruptions (Cretaceous–Eocene, about 66 Ma), about **5 lakh km²** over Maharashtra, Madhya Pradesh, Gujarat, and Karnataka.
- Black **regur** cotton soil is weathered trap basalt.

Trap layer	Depth fact
Upper	about **450 m**
Middle	about **1200 m**
Lower	about **150 m**
Intertrappean beds	Sediment **between** lava flows; **land / freshwater** fossils — **not** sea plants and animals



PYQ — UPPCS Prelims 2024, Q59

Which one of the following pairs (Deccan Trap – Peculiarity) is not correctly matched?

A. Depth of middle trap – approximately 1200 metres

B. Intertrappean beds – Fossils of sea plants and animals

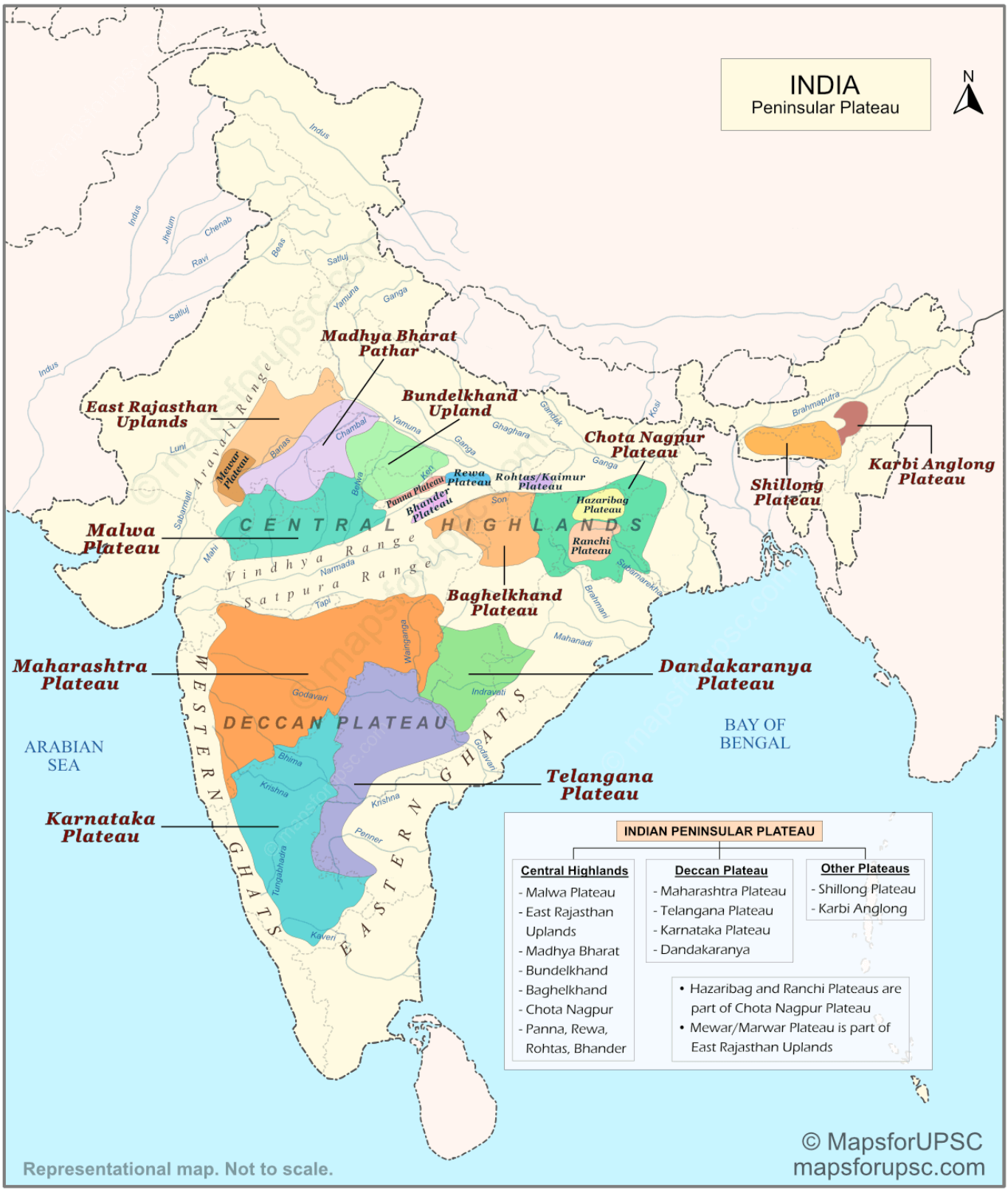
C. Depth of lower trap – approximately 150 metres

D. Depth of upper trap – approximately 450 metres

Show answer

Ans: B — Intertrappean = freshwater/land fossils. Depth figures A/C/D are the matched pairs.

Central Highlands and north-eastern plateaus



- The Central Highlands are the **northern** Peninsular Plateau, **north of the Narmada–Son**.
- **Marwar / Mewar (मेवाड़) Plateau** is **east** of the Aravalli (sandstone–shale–limestone, Banas toward Chambal). **Marwar Plain** is the Thar **west** of the Aravalli. Do not swap them.
- **Madhya Bharat Pathar** (Central Highland) is the Chambal basin with ravines / badlands.
- The **Malwa Plateau** is lava-covered and **north-sloping**. It has **dual drainage**: Narmada–Tapi–Mahi to the Arabian Sea, and Chambal–Betwa–Ken to the Yamuna / Bay of Bengal.

- The **Bundelkhand Plateau** lies in south-west Uttar Pradesh and Madhya Pradesh. It is **granite–gneiss**, drought-prone, and includes Jhansi (झांसी), Banda, and Chitrakoot (चित्रकूट). Betwa, Dhasan, and Ken cross it.
- The **Baghelkhand Plateau** lies **east** of Bundelkhand, **north of Maikal**. The Son bounds it on the north. It divides Son drainage from Mahanadi drainage.
- The **Chotanagpur Plateau** lies mainly in Jharkhand (with Odisha–West Bengal edges). Average height is about **700 m**. **Damodar** (दामोदर) flows in a **rift** through Gondwana coalfields.
- **Hazaribagh Plateau** lies north of Damodar. **Ranchi Plateau** lies south of Damodar.
- The **Rajmahal Hills** are the north-eastern basalt edge of Chotanagpur.
- The **Meghalaya Plateau** is Garo–Khasi–Jaintia. The **Malda / Garo–Rajmahal Gap** is a down-fault filled by Ganga–Brahmaputra alluvium.
- **Mikir (Karbi) Hills** are listed with this north-eastern block on the relief map.
- **Jharia** and **Bokaro** are the Damodar-rift coalfields. Iron and mica are the other Chotanagpur matchs.
- Chotanagpur is the easternmost unit of the west-to-east central-India hill sequence.

Other hills

- The **Rajmahal Hills** of Jharkhand are volcanic (**Rajmahal Trap**), older than the Deccan Trap.
- **Parasnath (1366 m)** is Jharkhand's highest peak and a Jain sacred hill on Chotanagpur.
- **Girnar (1117 m)** is Gujarat's usual highest point (Kathiawar).
- **Gudla Maina** (about 1270 m) is the usual Chhattisgarh high point on a Maikal extension.

1.4 Passes, Peaks & Sacred Geography

Mountain passes



PASS	CONNECTS	ALTITUDE (approx.)	STATE/UT
Karakoram Pass	India – China	5,540 m	Ladakh (J&K)
Shipki La	India – Tibet (China)	3,930 m	Himachal Pradesh
Nathu La	India – Tibet (China)	4,310 m	Sikkim
Lipulekh Pass	India – Tibet (China)	5,334 m	Uttarakhand
Niti Pass	India – Tibet (China)	5,033 m	Uttarakhand
Mana Pass	India – Tibet (China)	5,640 m	Uttarakhand
Bomdila	Not a pass (Town)	–	Arunachal Pradesh

UPPCS Trap: Lipulekh, Niti, Mana → Uttarakhand (NOT Ladakh/HP).

- UPPCS tests **pass ↔ State / UT**. The wrong state is the trap.

Pass	State / UT	Route / note	Trap
Burzil	J&K / Ladakh	Greater Himalaya (pass table)	—
Thaga La	Uttarakhand	Greater Himalaya with Niti / Lipulekh	—
Zoji La (ज़ोजिला)	Ladakh / J&K	Srinagar–Kargil–Leh (लेह)	Pair with Z-Morh tunnel
Banihal	J&K	Jammu–Srinagar; Jawahar Tunnel	Not a Tibet pass
Rohtang La	Himachal Pradesh	Kullu–Lahaul; Atal Tunnel below	Not Karakoram
Bara Lacha La	Himachal Pradesh	Manali–Leh	—
Kunzum La	Himachal Pradesh	Lahaul–Spiti	—
Shipki La	Himachal Pradesh	India–Tibet; Sutlej (सतलुज) valley	—
Niti Pass	**Uttarakhand**	Garhwal (गढ़वाल)–Tibet	2023 Q57 correct pair
Mana Pass	**Uttarakhand**	Near Badrinath (बद्रीनाथ)	**Not HP**
Lipulekh (लिपुलेख)	**Uttarakhand**	Kailash–Mansarovar; India–Nepal–China	**Not Ladakh**
Nathu La	Sikkim	India–Tibet trade (पण्याध्यक्ष) (reopened 2006)	—
Jelep La	Sikkim	Alternate to Nathu La	—
Bomdi La	Arunachal Pradesh	Near **Bomdila town**	Bomdila ≠ pass name
Sela Pass	Arunachal Pradesh	Tezpur–Tawang; **Sela Tunnel** (2024)	—
Bum La	Arunachal Pradesh	Tawang–Tibet	—
Pangsau Pass	Arunachal Pradesh	India–Myanmar (Stilwell Road)	—
Diphu / Aghil	Karakoram	Not Arunachal / not “Ladakh town” traps	2023 wrong options
Khardung La	Ladakh	Leh–Nubra road	High motorable pass; not a Tibet trade pass of the Nathu type

> **Logic:** Only Lipulekh–Ladakh is NOT matched **among common pass–state pairs. Nathu La–Sikkim and Shipki La–HP are correct.** Niti **and** Mana **are** Uttarakhand**, not Himachal.

PYQ — UPPCS Prelims 2025, Q55

Which of the following pairs are NOT correctly matched? (Pass) — (State/Union Territory)

1. Lipulekh — Ladakh
2. Nathu La — Sikkim
3. Bomdila — Arunachal Pradesh
4. Shipki La — Himachal Pradesh

A. Only 1 and 2

B. Only 2, 3 and 4

C. Only 1, 2 and 3

D. Only 1

Show answer

Ans: D — Only (1) is geographically wrong (Lipulekh = Uttarakhand). Key treats Bomdila–Arunachal as the accepted corridor pair.

State-wise highest peaks (match-list home)

State / UT	Peak	m	Range
Sikkim	Kanchenjunga	8586	Himadri
Uttarakhand	Nanda Devi	7816	Himadri
Himachal Pradesh	Reo Purgyil	6816	Kinnaur Himalaya
Arunachal Pradesh	Kangto	7090	Eastern Himalaya
J&K / Ladakh	Nun / Kun	7135 / 7077	Zanskar
West Bengal	Sandakphu	3636	Singalila
Nagaland	**Saramati**	3841	Naga Hills
Manipur	Tenipu / Mt Iso	2994	Manipur Hills
Mizoram	Phawngpui	2165	Mizo Hills
Meghalaya (मेघालय)	Shillong Peak	1965	Khasi
Kerala	**Anaimudi** (अनाइमुदी)	2695	Western Ghats
Tamil Nadu	**Doddabetta**	2637	Nilgiri
Karnataka	Mullayanagiri	1930	Western Ghats

State / UT	Peak	m	Range
Maharashtra	Kalsubai	1646	Western Ghats
Rajasthan (थार)	**Guru Shikhar**	1722	Aravalli
Madhya Pradesh	**Dhupgarh**	1350	Satpura
Andhra Pradesh	Arma Konda	~1680	Eastern Ghats
Odisha (ओडिशा)	Deomali	1672	Eastern Ghats
Gujarat	Girnar	1117	Kathiawar
Jharkhand	Parasnath	1366	Chotanagpur
Uttar Pradesh	Amsot (Sonbhadra)	~941	Kaimur
Chhattisgarh	Gudla Maina	~1270	Maikal extension

- Range highest (do not restudy the state table): Himalaya-in-India **Kanchenjunga 8586** · Karakoram **K2 8611** · Western Ghats **Anaimudi 2695** · Eastern Ghats **Jindhagada ~1690** · Aravalli **Guru Shikhar 1722** · Satpura **Dhupgarh 1350** · Nilgiri **Doddabetta 2637** · Naga **Saramati 3841**.

> **Logic:** Standard hill–state pairs: TN–Doddabetta, RJ–Guru Shikhar, NL–Saramati, MP–Dhupgarh. Older variants also used Kerala–Anaimudi and UK–Nanda Devi.

Sacred geography

Site	State	Hill / fact
Tirupati (Venkateswara)	Andhra Pradesh	**Tirumala / Mallamalla**, Eastern Ghats — **not Shevaroy**
Kedarnath (केदारनाथ)	Uttarakhand	Garhwal Himalaya; Mandakini (मंदाकिनी)
Badrinath	Uttarakhand	Garhwal Himalaya; Alaknanda (अलकनंदा)
Gangotri	Uttarakhand	Bhagirathi source glacier
Yamunotri	Uttarakhand	Yamuna source glacier
Amarnath	J&K	Lidder Valley (Greater Himalaya, not Pir Panjal)
Vaishno Devi	J&K	**Trikuta Hills**, Jammu
Sabarimala	Kerala	Western Ghats
Palitana (पालिताना)	Gujarat	Shatrunjaya Hills
Parasnath	Jharkhand	Chotanagpur
Girnar	Gujarat	Junagadh (जूनागढ़)
Ajmer Sharif	Rajasthan	Aravalli foothills
Chamundi	Karnataka	Mysuru

Site	State	Hill / fact
Konark (कोणार्क)	Odisha	**Coastal plain** (not a hill shrine)

> **Logic: Mallmalla** in options usually means **Tirumala Hills** (Eastern Ghats). **Shevaroy** = Yercaud (Tamil Nadu). **Biligiriranga** = Karnataka. **Javadhee** = Tamil Nadu.

1.5 Locational Framework

Latitude, longitude, area

- India lies entirely in the **northern** and **eastern** hemispheres.

Item	Note
Latitude	**8°4′ N – 37°6′ N** (about 29°; about 3214 km N–S)
Longitude	**68°7′ E – 97°25′ E** (about 29°; about 2933 km E–W)
Area	about **3.28 million km²** = **2.4%** of world land
Rank	Conventionally **7th** largest (after Russia, Canada, USA, China, Brazil, Australia)

- The **N–S and E–W degree spans are almost equal** (both about 29°). That is why NCERT maps show India as a near-square / diamond outline, not a long east–west strip.
- India is **not** wholly tropical. The Tropic of Cancer cuts the country, so the Himalayan north is subtropical to alpine.

Standard Meridian

Easy idea first: Earth spins once in 24 hours. Time is tied to **longitude** (north–south lines). Places farther **east** see sunrise **earlier**. India is wide enough for about **two hours** of natural solar difference, so one official line — the **Standard Meridian** — gives the whole country a **single clock time (IST)**.

- The Standard Meridian of India is **82°30′ E** (also written **82.5° E**). The classic place centre is **Mirzapur, Uttar Pradesh** (Prayagraj (प्रयागराज) / Allahabad (इलाहाबाद) belt). It is **not** Lucknow (लखनऊ), Delhi, or "exactly at Prayagraj city."
- **IST** is **GMT / UTC + 5 hours 30 minutes**. India keeps **one** time zone for the whole country (mainland + islands). CSIR–NPL maintains the official time signal.

Why 82°30′ E, not 80° or 85°?

Step	Easy fact
Earth	360° in 24 h means roughly **15°** of longitude per hour, or **1° ≈ 4 minutes**
Offset maths	82.5 ÷ 15 = **5.5 hours** = **5 h 30 min** ahead of GMT

Step	Easy fact
Why this line	It sits near the **middle** of India's longitude span (~68° E to ~97° E), so west (Gujarat) and east (Arunachal / Andaman) share one national clock

- Natural solar gap across India is about **2 hours** (roughly 29° × 4 min). **Gujarat** clocks feel "early" vs sun; **Arunachal / Andaman** feel "late" vs sun — but **legal time is still IST everywhere**.
- Sunrise is earlier in the **east** (Andaman / Arunachal) than in **Gujarat**. Do not reverse that.

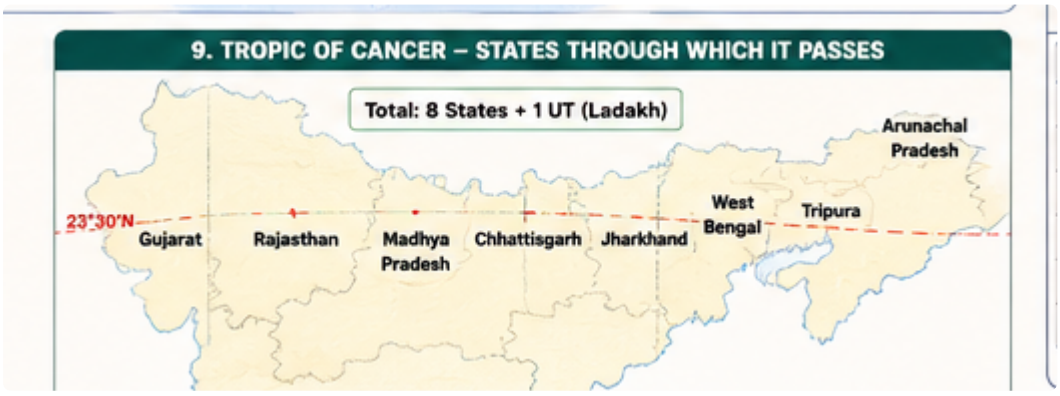
Where the meridian passes (north to south)

State	Place / belt fact	trap
Uttar Pradesh	**Mirzapur** (near Prayagraj belt)	Not Lucknow / Delhi; Tropic does **not** hit UP
Madhya Pradesh	Eastern MP / **Singrauli** belt	Not the whole of MP
Chhattisgarh	Central-east belt (**Bilaspur–Korba** class)	Not Jharkhand
Odisha	**Western** Odisha	Not the Bhubaneswar coast as the "IST city"
Andhra Pradesh	Northern / coastal AP strip	**Not Telangana** (TG lies west of 82.5° E)

- The IST meridian passes through **five states — UP, Madhya Pradesh, Chhattisgarh, Odisha, Andhra Pradesh**.
- It does **not** pass through **Uttarakhand, Jharkhand, Bihar, West Bengal, Telangana, Maharashtra, or Tamil Nadu**. Options that sneak in **Uttarakhand** or **Jharkhand** are classic wrong sets.
- Do **not** confuse this N–S **longitude** line with the Tropic of Cancer (an E–W **latitude** line). They are different. Both do cut through **Madhya Pradesh** and **Chhattisgarh**, so map stems can mix them.

> **Logic:** Place = **Mirzapur**. Offset = **+5:30**. States = **five** (UP–MP–CG–Odisha–AP). IST is **uniform** for all of India — never "only east of 82°30′ E."

Tropic of Cancer (23°30′ N)



Easy idea: This is a **latitude** line (runs east–west). South of it the sun can be overhead once a year; north of it (Himalaya belt) India is **not** purely tropical.

Order (west → east)	State on the Tropic
1	**Gujarat**
2	**Rajasthan**
3	**Madhya Pradesh**
4	**Chhattisgarh**
5	**Jharkhand**
6	**West Bengal**
7	**Tripura**
8	**Mizoram**

- **Eight states only.** Memorise the west-to-east chain; do not invent a ninth.
- It does **not** pass through **Uttar Pradesh**.
- It does **not** pass through **Ladakh** (Ladakh lies near 32°–36° N).
- It does **not** pass through Bihar, Odisha, Maharashtra, Karnataka, Kerala, Tamil Nadu, Punjab, Haryana, Delhi, Assam, Nagaland, Manipur, Arunachal, Sikkim, Himachal, Uttarakhand, Goa, Andhra Pradesh, or Telangana.
- NCERT says the Tropic passes through the **middle** of the country. That does **not** mean equal land area north and south of 23°30′ N: **more land lies north** of the Tropic.

> **Logic:** Rank = **7th**, not 6th. Area **2.4%**. Tropic through the **middle** matches the NCERT wording used in statement 3. India is **not** wholly tropical. UP is **not** on the Tropic.

PYQ — UPPCS Prelims 2022, Q35Logic: Stmt 1 tests rank (7th vs 6th). Stmt 2 tests area share. Stmt 3 is the NCERT “middle” wording (do not upgrade it to equal land area). Stmt 4 traps “wholly tropical.”

With reference to India, which of the following statements is/are correct?

1. India is the sixth largest country in the world.
2. India occupies about 2.4% of the total area of the world.
3. The Tropic of Cancer passes through the middle of the country dividing it into two latitudinal halves.
4. India lies completely in the tropical zone.

A. 2 and 3

B. 2 and 4

C. 3 and 4

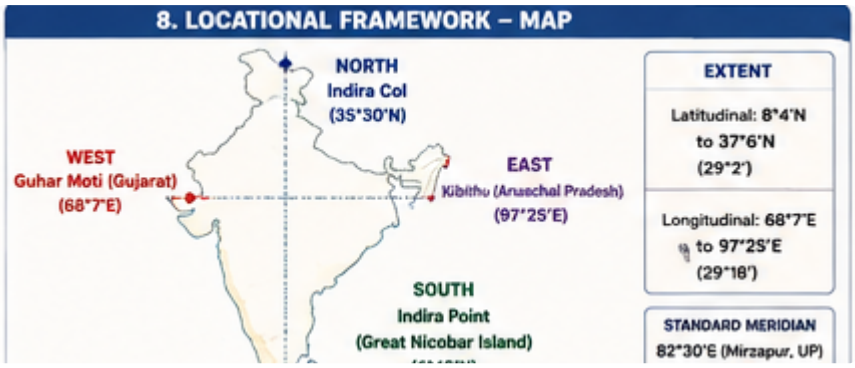
D. 1 and 2

Show answer

Ans: A — **2 and 3**. Rank is **7th**, not 6th (stmt 1 false). **2.4%** is true. Tropic through the middle is the NCERT fact (stmt 3) — not a claim of equal land area. India is **not** wholly tropical

(stmt 4). **B/C** keep wholly-tropical. **D** keeps the 6th-largest trap.

Extreme points



Extreme	Place	Easy fact
North	**Indira Col**, Siachen / Ladakh	High pass on the northern frontier — not a city
South (territory)	**Indira Point**, Great Nicobar	Southernmost **India** (islands included)
South (mainland)	**Kanyakumari** (Cape Comorin)	Southernmost **peninsula** only
East	**Kibithu**, Anjaw, Arunachal Pradesh	Near the Myanmar–China corner; not Walong as the usual key
West	**Guhar Moti** (also Ghuar Mota), Kutch, Gujarat	Westernmost village belt of Kachchh

- **Indira Col ≠ Indira Point.** Same first name, opposite ends of India.
- **Indira Point ≠ Kanyakumari.** Territory south vs mainland south.

Coastline

- Mainland plus islands is about **7516 km** (mainland about **6100 km** class figure). There are **nine** coastal states and two island Union Territories (**A&N** and **Lakshadweep**).

Rank (longest → shortest)	Coastal state
1	**Gujarat**
2	Andhra Pradesh
3	Tamil Nadu
4	Maharashtra
5	Kerala
6	Odisha
7	Karnataka

Rank (longest → shortest)	Coastal state
8	West Bengal
9	**Goa** (shortest state coast)

- Gujarat is longest because of the **Kachchh** and **Saurashtra** indentation (about **1600 km** class figure).
- The Andaman and Nicobar coastline is long (about **1962 km**) but it is **not** in the **state** ranking.
- **Telangana** is not a coastal state. **Puducherry** is a coastal UT with enclaves, not a “tenth coastal state” in this ranking.

> **Logic:** Gujarat **has the longest** state** mainland coastline — not Maharashtra, Andhra Pradesh, or Kerala.

